Beyond Accuracy: A Robustness Audit of Health Misinformation Classification Across CoAID and FakeHealth

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Introduction: Problem Description and Proposed Solution

Health misinformation creates a management and technical problem: organizations must triage items, justify warnings, and know when an automated system should defer. A high score is insufficient if a model learns publisher names, pandemic vocabulary, or dataset formatting instead of transferable indicators of information quality. Because misinformation varies across topics, platforms, definitions, and collection procedures (Baris Schlicht et al., 2023; Suarez-Lledo & Alvarez-Galvez, 2021), the practical question is whether a decision rule remains credible when source, time, dataset, and label construction change.

The proposed solution is a reproducible, human-centered audit pipeline combining sparse and transformer classifiers, controlled and cross-dataset tests, calibration and deferral, engagement ablations, narrative discovery, explanations, and structured review. It uses pandemic-specific, source-structured CoAID (Cui & Lee, 2020) and broader, expert-rated FakeHealth (Dai et al., 2020) as complementary rather than interchangeable tests. Throughout, *unreliable* denotes the harmonized dataset label, not an independent medical diagnosis.

Five questions guide the analysis: How well do sparse and transformer models classify each dataset? Do results survive source, temporal, and artifact controls? Do models transfer? Does engagement add information? Can calibration, narratives, explanations, and review identify cases that should not be trusted automatically? The contribution is a robustness and decision-support evaluation rather than a new architecture; losses under source control or transfer reveal boundaries on what the learned model can support.

Background: Literature Review and Industry Analysis

Prior research establishes both the public-health relevance of misinformation and the difficulty of defining it consistently. Its spread depends on messages, actors, networks, and interpreters (Wang et al., 2019), while reported prevalence varies by health topic and platform (Suarez-Lledo & Alvarez-Galvez, 2021). Current technical solutions include publisher rules, sparse text classifiers, and BERT-style contextual encoders (Devlin et al., 2019); industry responses add platform moderation and professional fact-checking. However, detection studies use labels ranging from expert judgments to publisher proxies, and binary outputs rarely communicate provenance, uncertainty, or transfer limits (Baris Schlicht et al., 2023). Public-health teams need rapid triage, platforms need defensible governance, and healthcare organizations face reputational harm from both misinformation and false alarms. This creates a need for an interpretable monitoring workflow that tests shortcuts and defers uncertain cases rather than treating internal accuracy as proof of medical truth.

Data Analysis and Testing of the Proposed Solution

The acquisition process pinned official revisions and retained inventories, checksums, native labels, parsing outcomes, duplicates, and exclusions. Of 8,271 parsed records, the primary cohort contains 3,055 CoAID articles and 2,137 FakeHealth items. CoAID claims lack comparable article bodies and were excluded. Other coded exclusions cover unavailable or placeholder pages, duplicates, conflicting labels, and bodies under 30 words; excluded rows remain auditable.

Standard manifests use approximately 70% training, 15% validation, and 15% testing while keeping connected duplicate families in one partition. Controlled evaluation follows the strongest feasible design. FakeHealth uses a canonical-publisher-disjoint split, ten repeated valid publisher allocations, and a hosting-domain sensitivity test. CoAID cannot satisfy the frozen publisher-disjoint feasibility gates because its 233 unreliable articles are concentrated among few sources; it therefore uses a temporal, family-disjoint fallback. These controlled conditions answer related but nonidentical questions and are not pooled.

Sparse candidates include word/bigram TF-IDF Logistic Regression and Linear SVM, publisher-, unit-, and length-only diagnostics, title/body ablations, and artifact-masked text. Each eligible sparse model compares 24 configurations on validation macro F1 before one test evaluation. Transformer benchmarks include pinned DistilBERT, ModernBERT, and BiomedBERT checkpoints, three seeds, early stopping, and a validation-only learning-rate audit. All were executed on Apple MPS. Transformer inputs are truncated to 512 tokens: only about 1.5% of CoAID exceeds this limit, compared with 81.8% of FakeHealth. The latter results therefore describe leading-window classification, not full-article transformer analysis.

Macro F1 is primary because both classes matter despite imbalance; unreliable-class precision, recall, F1, and precision-recall area are retained. Test uncertainty uses 1,000 family-, publisher-, or domain-grouped bootstrap resamples, as appropriate. Paired resampling compares models on identical records. Calibration methods and selective-prediction thresholds are chosen using separated validation subsets before test evaluation, following the principle that confidence should correspond to observed correctness (Guo et al., 2017). Transfer models train on one dataset and score the other's held-out test records after removing target domains seen during source training or validation.

FakeHealth engagement experiments compare text, engagement-ID counts, and combined models on the same complete cases. BERTopic clusters unreliable-labeled records; representatives are reviewed before a cluster is called a narrative (Grootendorst, 2022). A frozen 80-case audit samples ordinary and high-confidence errors, deferred cases, transfer failures, and narrative cases. Blind Stage A precedes explanations; Stage B confirms proposed error codes.

Findings: Classification and Robustness

Figure 1 compares the primary sparse baseline with the strongest transformer point estimate in each declared condition. Standard performance is high on CoAID, but the apparent strength is not source-independent. Logistic Regression falls from .857 macro F1 on the standard split to .675 temporally, while publisher-only Logistic Regression reaches .978 standard and .938 temporal. CoAID's language models therefore learn useful discrimination within the corpus, yet publisher identity remains a stronger shortcut than full text. The temporal transformer scores do not remove this concern because publishers still overlap across time.

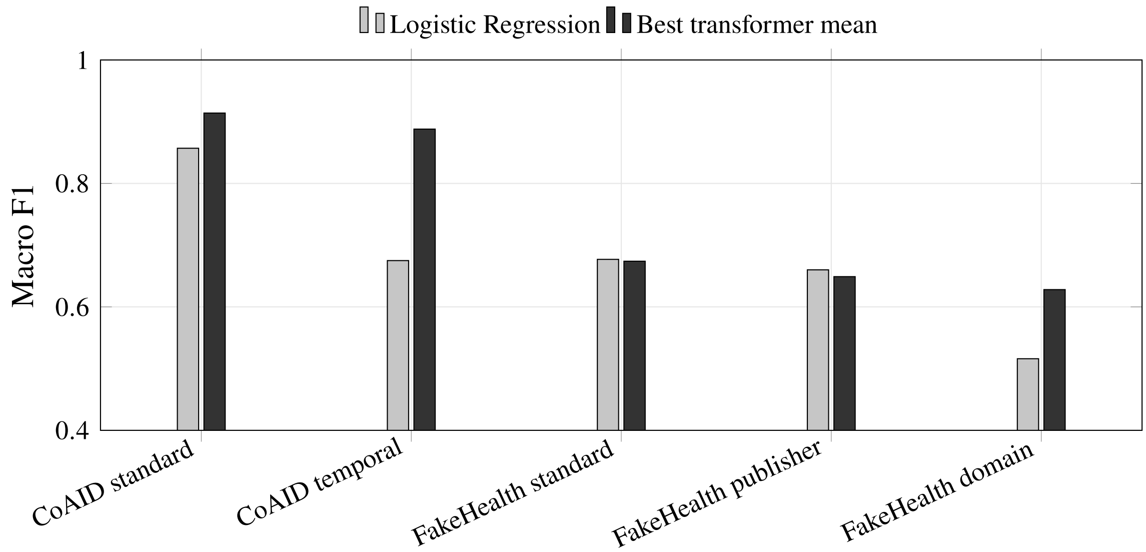

Figure 1

Primary macro-F1 results across standard and controlled tests. Transformer bars show the highest three-seed architecture mean within each condition, not a single universally selected model.

FakeHealth provides the stronger source-control result. Logistic Regression scores .677 standard and .660 with publishers held out; across ten valid publisher allocations its mean is .651 (SD = .062). The hosting-domain score is .516, but its 95% grouped-bootstrap interval (.266, .774) reflects only about five effective domain groups. Transformers do not clearly exceed the sparse baseline on ordinary FakeHealth evaluation: ModernBERT's best standard mean is .674. BiomedBERT reaches .628 in the domain test, yet architecture rank changes across conditions and the test remains small. The binary treatment of FakeHealth rating 3 is also consequential: excluding boundary-rating training items raises same-record Logistic Regression macro F1 from .735 to .791. Collectively, these results favor robustness profiles over a single leaderboard score.

Findings: Transfer, Decision Support, and Interpretation

Cross-dataset transfer supplies the clearest external-validity test. As Figure 2 shows, full-text Logistic Regression trained on CoAID scores .421 on 263 filtered FakeHealth test records, compared with .686 for the target-trained same-record baseline. Training on FakeHealth and transferring to 161 CoAID records is worse: .309 versus .874. These losses reject portability of the fitted decision rules. They do not establish that either dataset is medically correct; topic, publication period, text unit, publisher composition, and label provenance all change simultaneously.

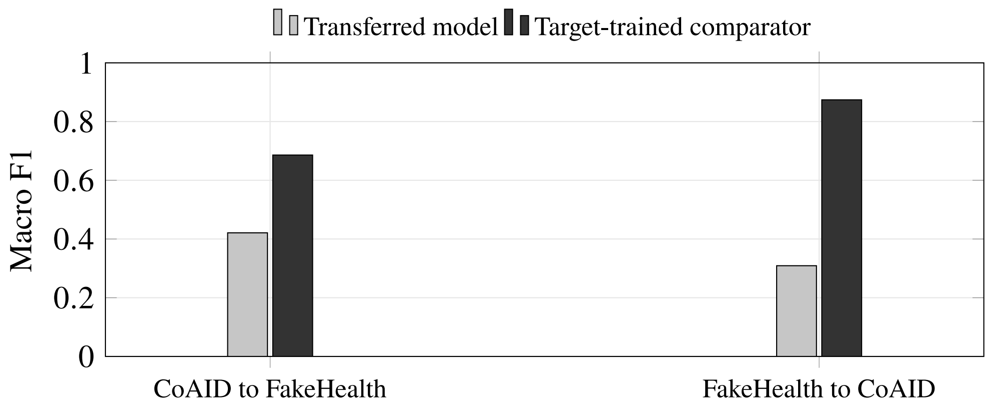

Figure 2

Full-text Logistic Regression transfer versus a target-trained comparator evaluated on the same filtered target records.

Decision-support analyses are useful but do not support autonomous adjudication. Platt-calibrated CoAID Logistic Regression reduces selective risk from 3.9% to 0.5% at 84.5% test coverage, but retained cases are almost entirely reliable-side predictions; unreliable-side coverage is zero. FakeHealth coverage is 81.6%, while risk declines from 30.0% to 24.5%, still too high for automatic health-information decisions. Deferral can prioritize review, not confer medical validity.

On matched FakeHealth records, adding log-transformed engagement counts changes macro F1 from .667 to .677 standard and from .687 to .700 publisher-disjoint. Both paired confidence intervals include zero, so incremental benefit is not established. Narrative discovery finds no size-eligible CoAID cluster. Four FakeHealth clusters undergo human review; three are retained, covering 321 of 720 unreliable-labeled items and describing cancer advances, women's reproductive or menopause treatments, and cardiac procedures or devices. This 44.6% is corpus prevalence, not internet prevalence. The 80-case qualitative audit identifies topic shortcuts and medical-terminology errors frequently within its deliberately enriched sample. Because Stage B proposals were generated and then approved without overrides, these codes are exploratory assisted-review evidence rather than independent agreement data.

Recommendations and Implications for Practice

Public-health agencies should use the pipeline to rank items for trained or clinical review while publishing correction evidence and uncertainty. Platforms should monitor performance by publisher, topic, time, and calibration; preserve appeal and audit trails; and avoid source identity as a substitute for claim-level assessment. Healthcare organizations should monitor emerging narratives and reputational exposure without treating an automated label as a medical judgment. Across settings, interfaces should present risk, confidence, source and temporal context, sensitive terms, narrative, and a defer option. Engagement volume should inform reach only after exposure and missingness are understood.

Discussion

Classification performance is conditional on dataset construction. CoAID achieves excellent internal scores while publisher identity remains more predictive than language. FakeHealth's publisher-disjoint result is more encouraging, but domain uncertainty and truncation remain substantial. Failed bidirectional transfer confirms that these reliability labels do not define a stable universal construct, consistent with warnings about heterogeneous topics and labeling strategies (Baris Schlicht et al., 2023). The appropriate product is therefore review support, not an automated truth engine.

Limitations. CoAID is COVID-specific and source-structured; FakeHealth uses health-news review scores. Only 233 CoAID primary articles are unreliable, preventing publisher-disjoint testing. FakeHealth transformers see 512 leading tokens although 81.8% of items are longer. The domain test has few effective groups, neural comparisons use three seeds, engagement is incomplete, and one reviewer completed the qualitative stages. Topics are descriptive clusters, not validated claims.

Future work. Evaluate contemporary, independently annotated data with trained or clinical reviewers; obtain enough minority examples and publishers for source-disjoint testing; compare models on matched 512-token inputs before testing chunking or longer contexts; seed initialization and archive against a clean commit; and independently recode qualitative cases with a second reviewer.

Conclusion

The project succeeds as a robustness audit. Models discriminate dataset-defined reliability, but strong internal performance coexists with publisher shortcuts, domain sensitivity, and transfer failure. Transformers improve CoAID point estimates without a clear FakeHealth gain; engagement adds no established value. Calibration, narratives, explanations, and deferral instead provide review context. The defensible path is human-centered monitoring with transparent uncertainty, not autonomous medical fact-checking.

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